# Executive overview

INTERNAL AI APPLICATION

PRODUCTION

## Settlement Failure Early Warning

AI-INT-2026-041 | Model v2.3.1

| 0.95 | 0.91 | Medium | Daily |
|---|---|---|---|
| ROC-AUC | Precision | Residual risk | Metric checks |

### BUSINESS OBJECTIVE

Identify likely T+1 settlement exceptions early enough for operations analysts to prioritize investigations. Outputs are advisory and cannot release, block, or alter settlement instructions without a human decision.

### DECISION WORKFLOW

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| Transaction / attributes | Risk score / & alert | Analyst / review | Manual / action |

### LATEST OVERSIGHT FINDINGS

**WATCH ITEM**

A September batch-feed delay triggered manual fallback. Incident closed; drift PSI is 0.14 against a review threshold of 0.20.

# Structured inventory | ownership, data and controls

## CANONICAL INVENTORY FIELDS

22 mapped fields • values are fictional • explicit IDs support extraction testing

**01 / Submission ID**
AI-INT-2026-041

**02 / Legal entity**
Example Post-Trade Operations (fictional)

**03 / Application name**
Settlement Failure Early Warning (SFEW)

**04 / Business use case**
T+1 settlement exception prediction

**05 / Business owner**
Maya Chen, Director, Post-Trade Analytics

**06 / Technical owner**
Alex Rivera, Senior ML Engineer

**07 / Model type**
Gradient-boosted trees (XGBoost)

**08 / AI provider**
Internally developed; no external inference API

**09 / Model version**
2.3.1

**10 / Lifecycle status**
Production

**11 / First deployment**
2026-07-15

**12 / Input data**
Transaction metadata + exception history

**13 / Data sensitivity**
Confidential, transaction-level data

**14 / Financial criticality**
High; advisory operations prioritization

**15 / Human oversight**
Analyst approves every escalation

**16 / Validation status**
Passed independent validation on 2026-10-20

**17 / Performance**
Precision 0.91; recall 0.86; ROC-AUC 0.95

**18 / Risk rating**
Inherent HIGH; residual MEDIUM

**19 / Incident history**
1 closed incident; see page 4

**20 / Monitoring frequency**
Daily metrics; monthly drift review

**21 / Fallback process**
Manual rules-based exception queue

**22 / Next periodic review**
2027-01-20

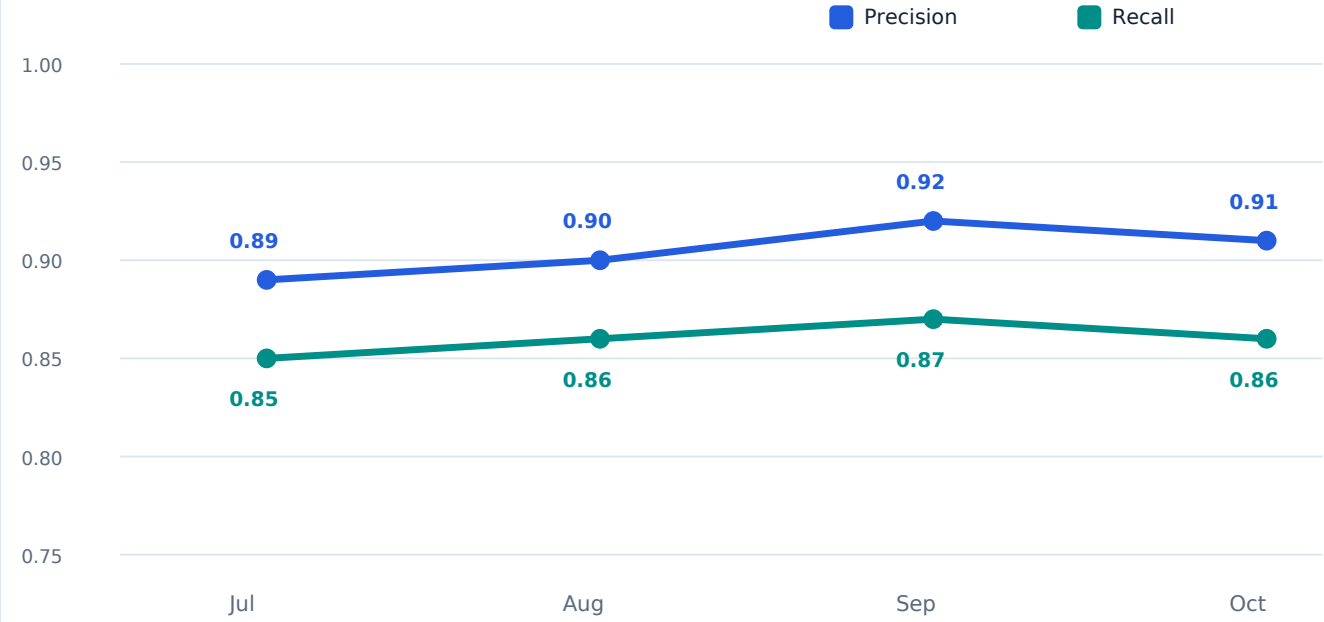
**DATA-HANDLING NOTE**

Synthetic data only in this sample. Real transaction-level inputs would require approved access controls, retention limits, and audit logging.

# Model validation | metrics and drift

## MONTHLY MODEL PERFORMANCE

Holdout/production shadow sample; monthly figures are synthetic.




## VALIDATION RESULTS

| Measure | Observed | Threshold | Result |
|---|---|---|---|
| Precision | 0.91 | >= 0.85 | **Pass** |
| Recall | 0.86 | >= 0.80 | **Pass** |
| ROC-AUC | 0.95 | >= 0.90 | **Pass** |
| Feature drift (PSI) | 0.14 | Review >= 0.20 | **Below trigger** |

### VALIDATION EVIDENCE

Independent review completed 2026-10-20 using a 30,000-record retrospective test set. No material threshold breach; next review scheduled for 2027-01-20.

# Governance | risk register and incident review

## MATERIAL RISK & CONTROL REGISTER

| Risk | Mitigation / control | Residual | Control owner |
|---|---|---|---|
| Model drift | Monthly PSI + outcome monitoring | **Medium** | Model Risk |
| Data leakage | RBAC, encryption, access audit | **Low** | InfoSec |
| False alerts | Human review before escalation | **Medium** | Operations |
| Operational resilience | Manual queue + batch feed checks | **Medium** | Platform Ops |

## INCIDENT REGISTER

**INC-2026-09-12-01 • Closed 2026-09-13**

Category: Data pipeline / batch feed latency

Impact: 42-minute delay in risk-score availability; 0 settlement actions affected.

Response: Manual exception queue activated; delayed alerts were reprocessed.

Preventive action: Feed freshness alarm reduced from 30 to 10 minutes.

## MONITORING & OVERSIGHT CADENCE

| | | |
|---|---|---|
| **Every business day** | Precision/recall proxies, alert volumes, feed freshness | Operations + ML Ops |
| **Monthly** | PSI drift review, false-positive sampling, access logs | Model Risk |
| **Quarterly** | Control attestation and material change review | AI Governance |

Decision record: Residual MEDIUM risk accepted for advisory use only; human approval is mandatory. Model changes require revalidation before release.